



**WIF3009**

# **Python for Scientific Computing**

Statistical analysis using Python



# Statistics Basics

# Statistics vs Statistical inference



## Statistics

- The science of collecting, organizing, analyzing and interpreting data in order to make decision in the presence of uncertainty

## Statistical Inference

- Make use of information from a sample to draw conclusions about the population from which the sample was taken.

# Statistics (what is ?)

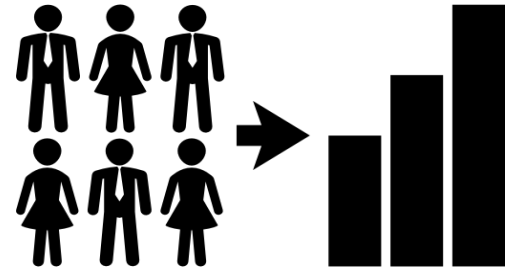
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- **Statistics**, in general, is the method of collection of data, tabulation, and **interpretation of numerical data**.
- It is an area of applied mathematics concerned with data collection **analysis, interpretation, and presentation**.
- With statistics, we can see **how data can be used** to solve complex problems.



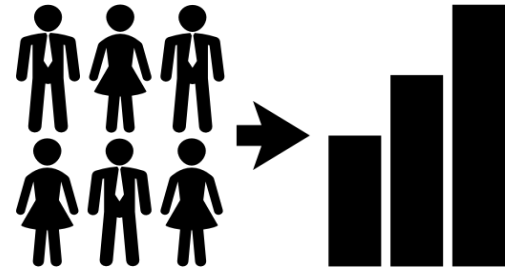
# Statistics – Term (Population)

- Population is the **set representing all measurements** of interest to the investigator.
- A population is **any entire collection** of people, animals, plants or things from which we may collect data.
- It is the **entire group** we are interested in, which we wish to describe or draw conclusions about.
- In order to make any generalizations about a population, a sample, that is meant to be representative of the population, is often studied.



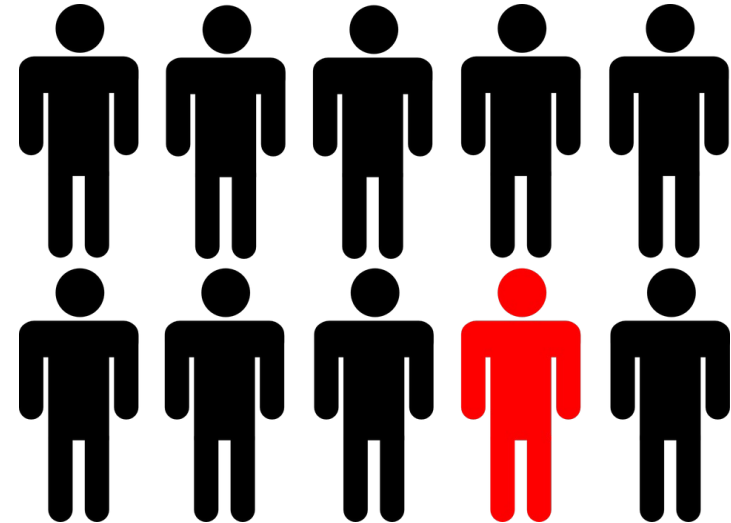
# Statistics – Term (Sample)

- A **sample** is a **subset** of measurements selected from the population of interest.
- The sample should be **representative of the population**.
- A sample could be the whole population,
  - e.g. Malaysia Census in ideal situations.
- In many cases the population is conceptual,
  - e.g. Daily yields of a power plant for next 2 years.
- In most cases measuring the **whole population is too costly**, unnecessary or impossible.
- There are **many possible samples** when sample is a subset of the whole population.



# Statistics – Example ( The Prime Minister Approval Rating )

- A New Straits Time pool of 1500 adult Malaysian conducted on January 15, 2004 shows that :
  - 618 said "approve".
  - 702 said "disapprove" .
  - 180 either declined to answer or has no opinion.
- Population:
  - 23-plus million adult Malaysian.
- Sample:
  - 1500 interviewed.



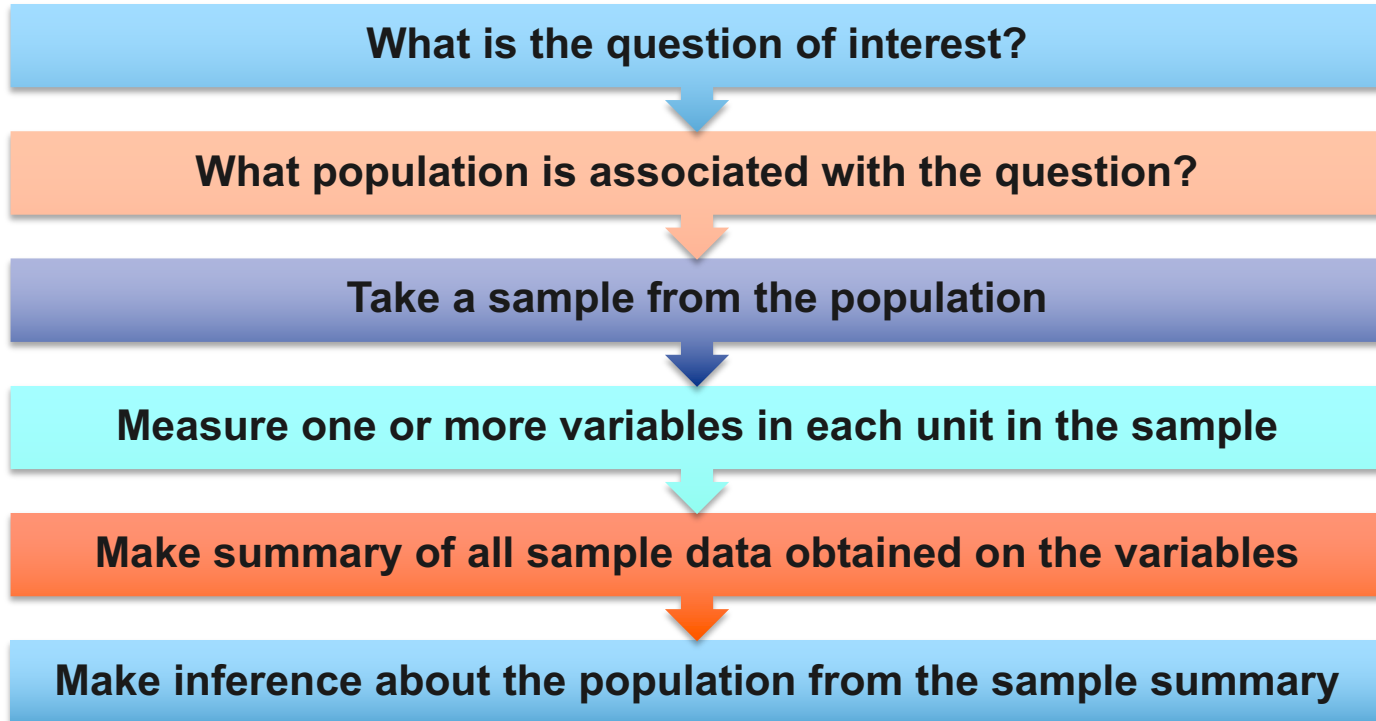
## Statistics – Inference

*“Inference is the act or process of **deriving logical conclusions** from premises known or assumed to be true. The **conclusion** drawn is also called an idiomatic.”*

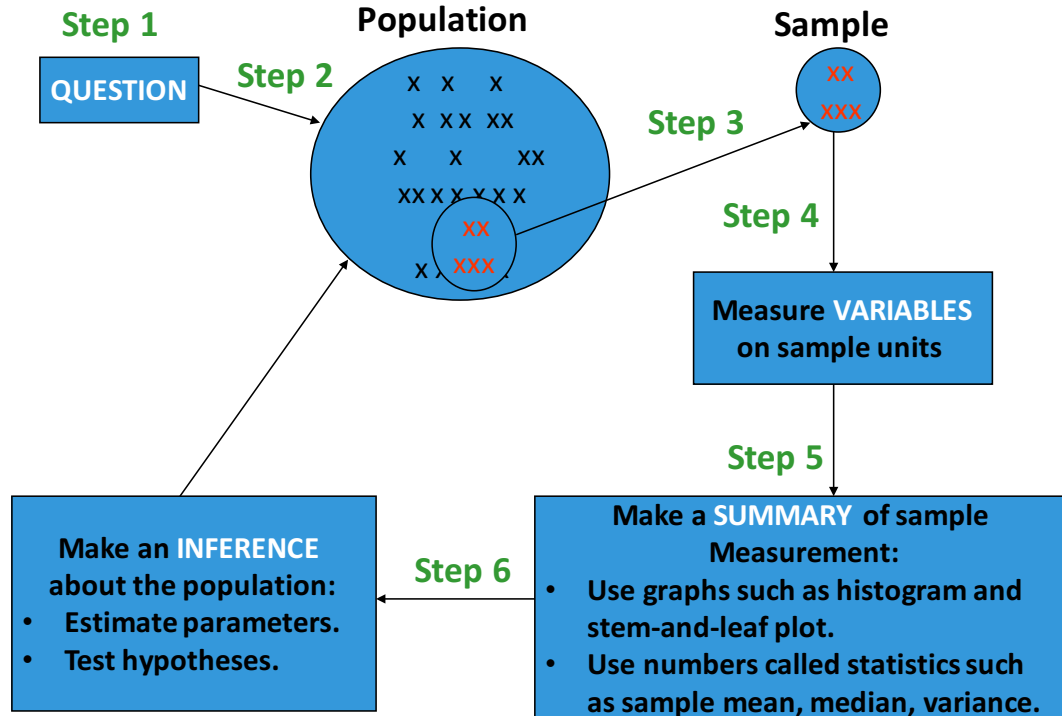




# Statistics – The Six Step Inference Process



# Statistics – The Six Step Inference Process



## Statistics –Example ( Pollution at an Oil Refinery )

- Environmental Protection Agency (**EPA**) has accused BHP Oil Company of **violation environmental regulations** at its refinery located in Sarawak in 2021
- The regulations state that the average petroleum leaked into the ground at the refinery **must not exceed 100 gallons per day** during any calendar year.
- Fine for violating the regulations is RM1,000,000.
- EPA regulators visited the refinery on eight days in December, 2001 and measured the petroleum leaked in gallons as:
  - 110, 96, 104, 101, 87, 99, 116, 108.



# Statistics –Example ( Pollution at an Oil Refinery )

- The **Six Step** Inference Process:
  - **Question:** Does average leakage exceed 100 gallons per day?
  - **Population:** Every single day of that year 2001.
  - **Sample:** Measurements on those 8 days in December.
  - **Variable:** Leakage in gallons.
  - **Summary:** Average leakage=102.625 gallon/day.
  - **Inference:** The average leakage for the year 2001 exceeds 100 gallons. The company **should be fined**.



# Python - Scipy

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- SciPy is a **data-processing library** that is made to compete with its rivalries such as MATLAB, Octave, R-Lab, etc.
- It has many easy-to-use functions that help to solve problems like **integration**, **interpolation**, **optimization**, **linear algebra**, and **statistics**.
- Benefit:
  - Makes a strong programming language available for developing complex programs and applications.
- How to get it:
  - `pip install scipy`



# Statistics - Interpreting Graphs (Location and Spreads)



**Mound shaped and symmetric (mirror images)**



**Skewed right: A few unusually large measurements**



**Skewed left: A few unusually small measurements**



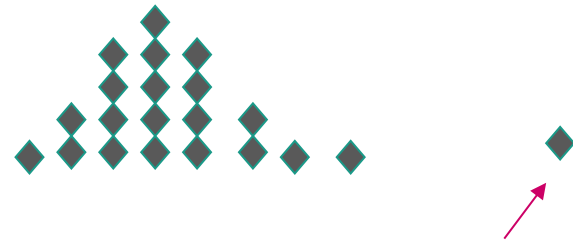
**Bimodal: Two local peaks**

# Statistics - Interpreting Graphs (Outliers)

- Are there any strange or unusual measurements that stand out in the data set?
- **Outlier** is an observation point that is **distant from other observations**.
- An outlier may be **due to variability** in the measurement or it may indicate **experimental error**. The latter are sometimes excluded from the data set.
- Outlier **can cause serious problems** in statistical analyses.



No Outliers

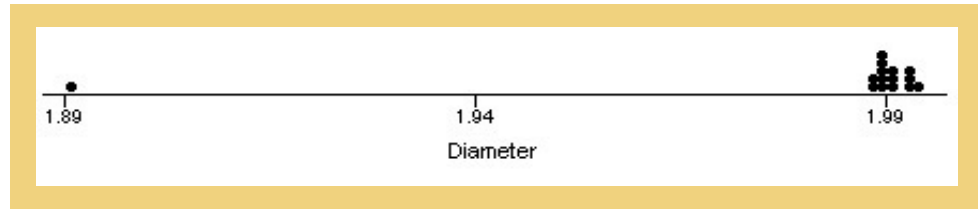


Outlier

## Statistics - Example (Outliers)

- A quality control process measures the diameter of a gear being made by a machine (cm).
- The technician records 15 diameters, but inadvertently makes a typing mistake on the second entry.

[1.991, 1.891, 1.991, 1.988, 1.993, 1.989, 1.990, 1.988  
1.988, 1.993, 1.991, 1.989, 1.989, 1.993, 1.990, 1.994]



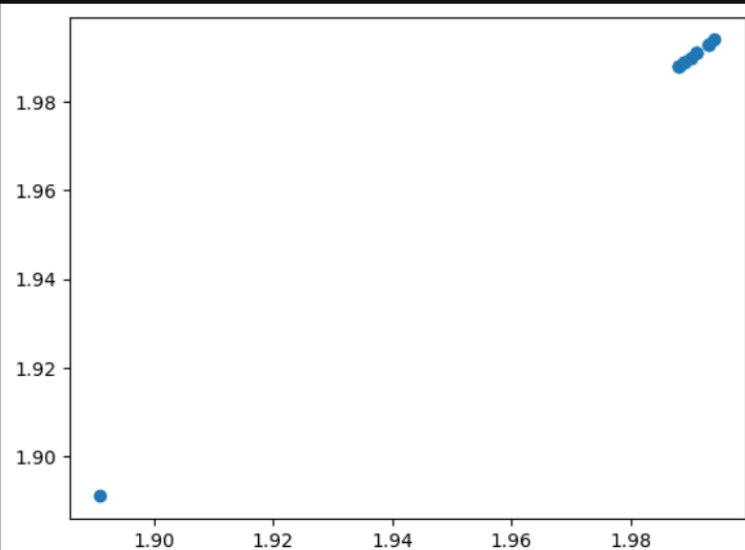


# Statistics - Example (Outliers)

```
import matplotlib.pyplot as plt
```

```
plt.scatter(data,data)
```

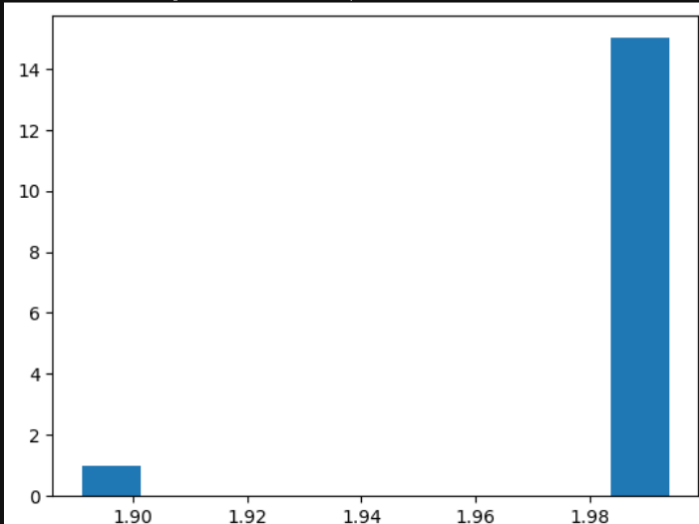
```
<matplotlib.collections.PathCollection at 0x2115872f850>
```



```
import matplotlib.pyplot as plt
```

```
plt.hist(data)
```

```
(array([ 1.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0., 15.]),  
 array([1.891 , 1.9013, 1.9116, 1.9219, 1.9322, 1.9425, 1.9528, 1.9631,  
        1.9734, 1.9837, 1.994 ]),  
 <BarContainer object of 10 artists>)
```



# Statistics - Example

- The ages of 50 tenured faculty at a public university.

34 48 70 63 52 52 35 50 37 43 53 43 52 44  
42 31 36 48 43 26 58 62 49 34 48 53 39 45  
34 59 34 66 40 59 36 41 35 36 62 34 38 28  
43 50 30 43 32 44 58 53

- Intervals = 6 was chosen.
- Minimum class width =  $(70 - 26)/6 = \underline{7.33}$
- Convenient class width = 8
- Use 6 classes of length 8, starting at 25.

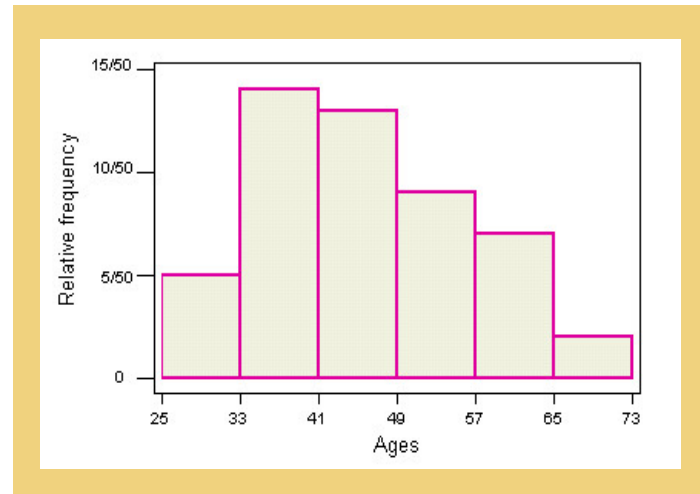
## Statistics – Example [cont]

- The statistics table of the ages of 50 tenured faculty at a public university.

| Age        | Tally          | Frequency | Relative Frequency | Percent |
|------------|----------------|-----------|--------------------|---------|
| 25 to < 33 | 1111           | 5         | $5/50 = 0.10$      | 10%     |
| 33 to < 41 | 1111 1111 1111 | 14        | $14/50 = 0.28$     | 28%     |
| 41 to < 49 | 1111 1111 111  | 13        | $13/50 = 0.26$     | 26%     |
| 49 to < 57 | 1111 1111      | 9         | $9/50 = 0.18$      | 18%     |
| 57 to < 65 | 1111 11        | 7         | $7/50 = 0.14$      | 14%     |
| 65 to < 73 | 11             | 2         | $2/50 = 0.04$      | 4%      |

## Statistics – Example [cont]

- Describing the distributions of the ages of 50 tenured faculty at a public university.
- Shape: Skewed right
- Outliers: No
- What proportion of the tenured faculty are younger than 41?
  - $(14 + 5)/50 = 19/50 = \underline{0.38}$
- What is the probability that a randomly selected faculty member is 49 or older?
  - $(9 + 7 + 2)/50 = 18/50 = \underline{0.36}$



# Statistics – Mean



- The mean of a set of measurements is the sum of the measurements divided by the total number of measurements.

$$\bar{x} = \frac{\sum x_i}{n}$$

where  $n$  = number of measurements

$\sum x_i$  = sum of all the measurement s

- If we were able to enumerate the whole population, the population mean would be called  $\mu$  (the Greek letter “mu”).

# Statistics – Median

- The median of a set of measurements is the middle measurement when the measurements are ranked from smallest to largest.
- The position of median is  $.5(n + 1)$  once the measurements have been ordered:

- Example:

The set: 2, 4, 9, 8, 6, 5, 3       $n = 7$

Sort: 2, 3, 4, 5, 6, 8, 9

Position:  $.5(n + 1) = .5(7 + 1) = 4^{\text{th}}$

Median = 4<sup>th</sup> largest measurement

The set: 2, 4, 9, 8, 6, 5       $n = 6$

Sort: 2, 4, 5, 6, 8, 9

Position:  $.5(n + 1) = .5(6 + 1) = 3.5^{\text{th}}$

Median =  $(5 + 6)/2 = 5.5$  — average of the 3<sup>rd</sup> and 4<sup>th</sup> measurements

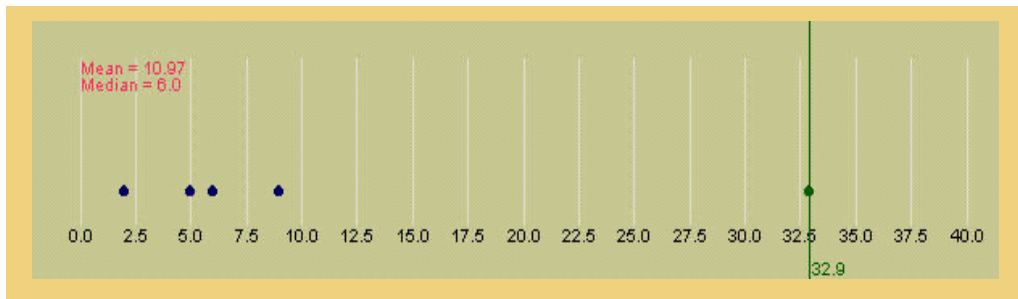
# Statistics – Mode



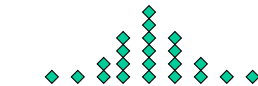
- The mode is the measurement which occurs most frequently.
- The set: 2, 4, 9, 8, 8, 5, 3
  - The mode is 8, which occurs twice
- The set: 2, 2, 9, 8, 8, 5, 3
  - There are two modes—8 and 2 (bimodal)
- The set: 2, 4, 9, 8, 5, 3
  - There is no mode (each value is unique).

# Statistics – Extreme Values

- The **mean** is more easily **affected** by extremely large or small values than the median.



- The **median** is often used as a measure of center when the **distribution is skewed**.



Symmetric: Mean = Median



Skewed right: Mean > Median

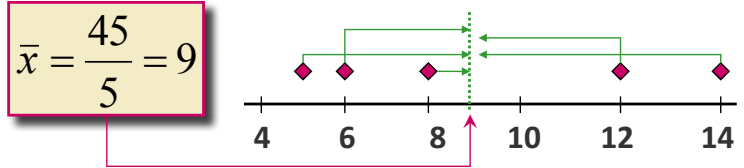


Skewed left: Mean < Median



# Statistics – Variance

- The variance is **measure of variability**.
- Measures the **average deviation** of the measurements about their mean.
- E.g. Flower petals: 5, 12, 6, 8, 14

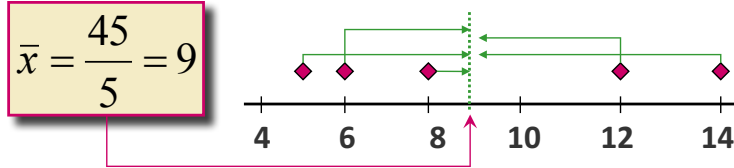


- The variance of a sample of  $n$  measurements is the sum of the squared deviations of the measurements about their mean, divided by  $(n - 1)$ .

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

# Statistics – Variance

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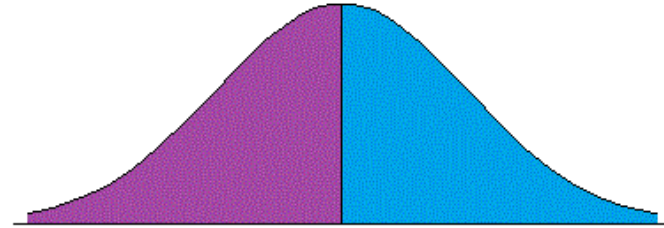
# Statistics – Standard Deviation

- In calculating the variance, we squared all of the deviations, and in doing so **changed the scale** of the measurements.
- To return this measure of variability to the original units of measure, calculate the standard deviation, the **positive square root of the variance**.

Population standard deviation :  $\sigma = \sqrt{\sigma^2}$

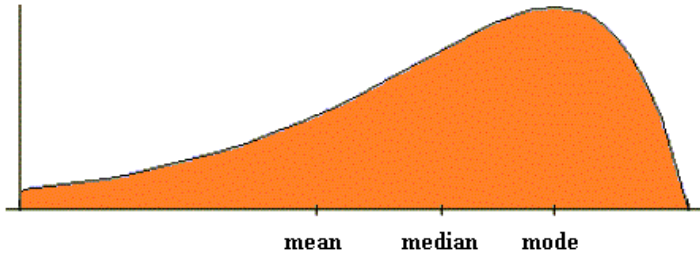
Sample standard deviation :  $s = \sqrt{s^2}$

# Statistics – Distribution

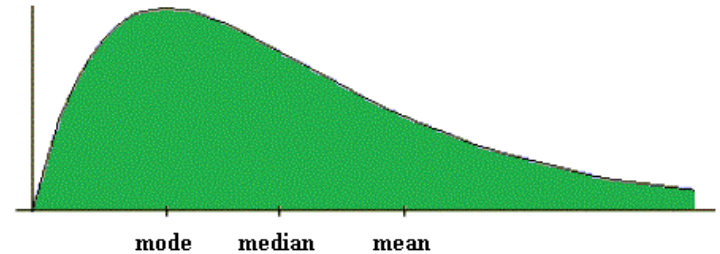


mean = median = mode

**Symmetric**

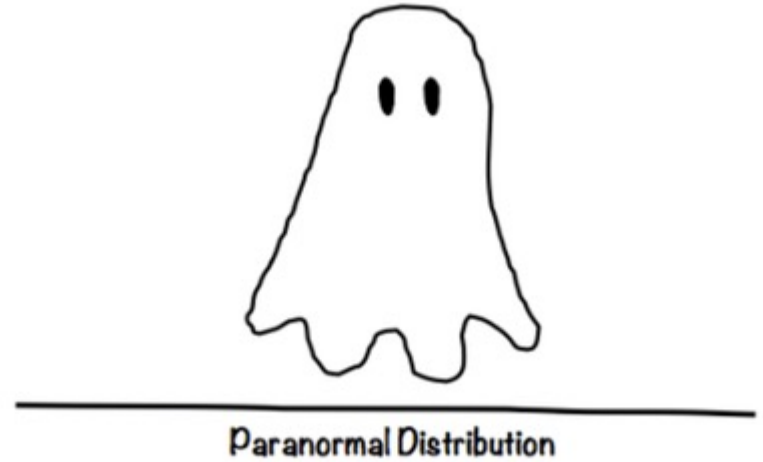
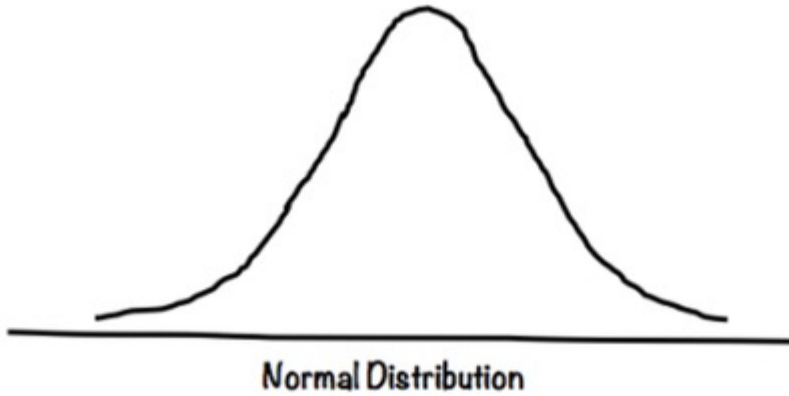


**Skewed left**



**Skewed right**

# Statistics – Distribution



# Statistics – In Scipy?

- Easily calculate in Scipy using the `.describe()` function

```
import scipy as sc
```

```
data = [1.991, 1.891, 1.991, 1.988, 1.993, 1.989, 1.990, 1.988, 1.988, 1.993, 1.991, 1.989, 1.989, 1.993, 1.991, 1.991]
```

```
sc.stats.describe(data)
```

```
DescribeResult(nobs=16, minmax=(1.891, 1.994), mean=1.98425, variance=0.0006222000000000007, skewness=-3.576346924913397, kurtosis=10.895883903830013)
```

**Kurtosis** => Indicates how much data resides in the tails of a data distribution

```
sc.ndimage.median(data)
```

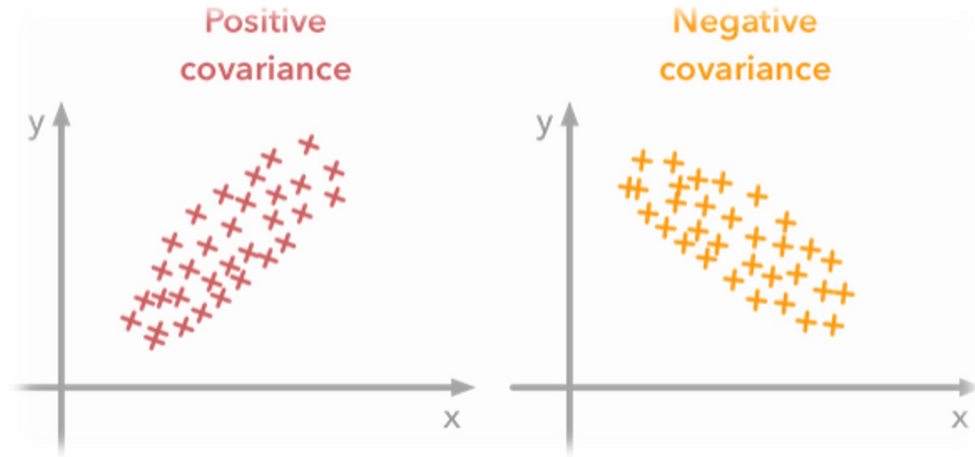
```
1.99
```



# **Statistical Analysis (SA) - Basics**

# SA – Covariance

- The covariance of two variables  $x$  and  $y$  in a data set **measures how the two are linearly related** (strength of the correlation between two).
- A **positive** covariance would indicate a positive linear relationship between the variables.
- A **negative** covariance would indicate the opposite.





# SA – Covariance



- In python, you can use ***numpy.cov()*** function
  - If  $\text{COV}(x_i, x_j) = 0$  then variables are uncorrelated
  - If  $\text{COV}(x_i, x_j) > 0$  then variables positively correlated
  - If  $\text{COV}(x_i, x_j) < 0$  then variables negatively correlated

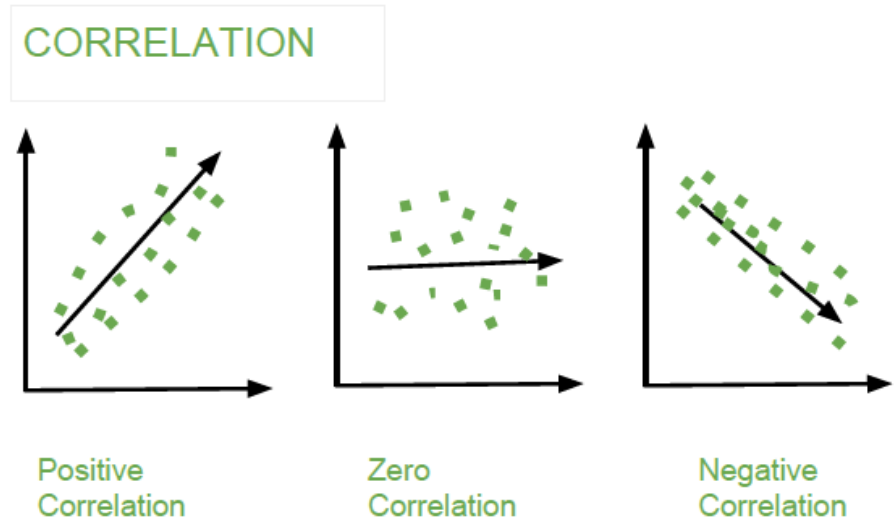
```
import numpy as np
```

```
np.cov(data)
```

```
array(0.0006222)
```

# SA – Correlation Coefficient

- A number that is a measure of the strength and direction of the correlation between two variables.
- The correlation is closely related to the covariance.
- Defined to ensure that its value is always between -1 and +1
- It is a normalized measurement of how the two are linearly related.



# SA – Correlation Coefficient



- Can use scipy to calculate

- Person's  $r$  (-1 to 1)
- Spearman's  $\rho$  (-1 to 1)
- Kendall's  $\tau$  (0 to 1)

- Example data:

```
x = np.array([1,3,5,7,8,9, 10, 15])
```

```
y = np.array([10, 20, 30, 40, 50, 60, 70, 80])
```

# SA – Correlation Coefficient



```
sc.stats.pearsonr(x,y)
```

```
[72]:
```

```
PearsonRResult(statistic=0.974894414261588, pvalue=3.8818392694035655e-05)
```

```
sc.stats.spearmanr(x,y)
```

```
[73]:
```

```
SignificanceResult(statistic=1.0, pvalue=0.0)
```

```
sc.stats.kendalltau(x,y)
```

```
[74]:
```

```
SignificanceResult(statistic=0.9999999999999998, pvalue=4.96031746031746e-05)
```

## SA – Correlation Coefficient

- What if you have a dataframe with multiple columns?
- Pandas can easily calculate the corr-coeff for you.
- You need to use the `.corr()` function

```
import pandas as pd  
  
brain = pd.read_csv('data/brain_size.csv', sep=';')  
brain.iloc[:, 1:].corr(method='pearson')
```

[76]:

|           | FSIQ      | VIQ       | PIQ       | Weight    | Height    | MRI_Cou |
|-----------|-----------|-----------|-----------|-----------|-----------|---------|
| FSIQ      | 1.000000  | 0.946639  | 0.934125  | -0.087655 | -0.092116 | 0.3576  |
| VIQ       | 0.946639  | 1.000000  | 0.778135  | -0.109010 | -0.077425 | 0.3374  |
| PIQ       | 0.934125  | 0.778135  | 1.000000  | -0.031881 | -0.082455 | 0.3868  |
| Weight    | -0.087655 | -0.109010 | -0.031881 | 1.000000  | 0.684494  | 0.4858  |
| Height    | -0.092116 | -0.077425 | -0.082455 | 0.684494  | 1.000000  | 0.5994  |
| MRI_Count | 0.357641  | 0.337478  | 0.386817  | 0.485832  | 0.599419  | 1.0000  |

# SA – Correlation Coefficient

- Use seaborn package to create correlation viz.
- Just *pip install seaborn* (another viz library)

```
import pandas as pd
import seaborn as sns

brain = pd.read_csv('data/brain_size.csv', sep=';')
corr = brain.iloc[:, 1:].corr(method='pearson')

plt.figure(figsize=(10, 8), dpi = 500)
sns.heatmap(corr, annot=True, fmt=".2f", linewidth=.5)
plt.show()
```

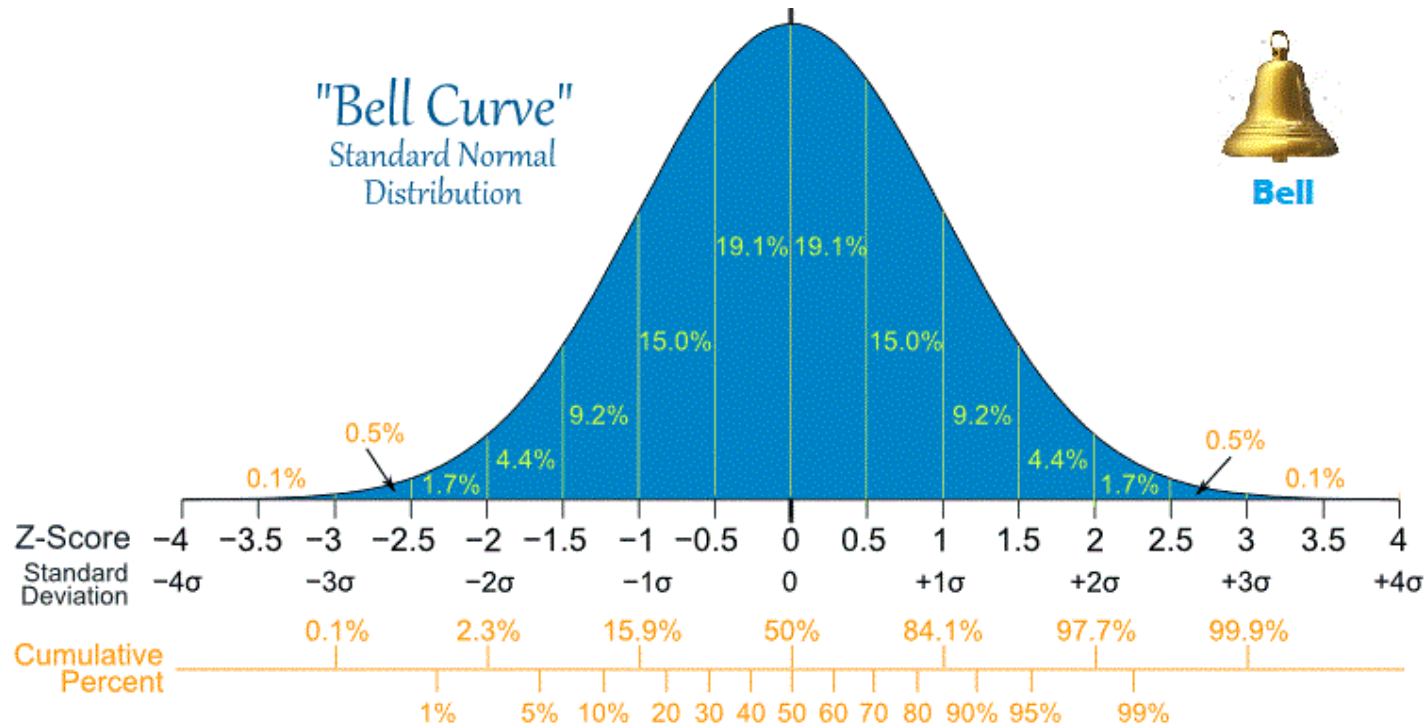


# SA – Statistical Distribution



- One of the most well-known distributions, also known as the bell-shaped curve.
- Its overall shape, when the data are organized in graph form, is a symmetric bell-shape.
- Around **68%** of the data are centered around the mean
- Moving farther out on either side of the mean, you find fewer and fewer values
- Examples of Normal Distribution:
  - Heights of people
  - Blood pressure
  - Marks on a test
- Other distributions such as:
  - Binomial
  - Poisson

# SA – Statistical Distribution





# SA – Statistical Inference



- A **statistical inference** aims at learning characteristics of the population from a sample
  - The population characteristics are *parameters* and sample characteristics are *statistics*.
- A **statistical model** is a mathematical formulations that describe relationships between random variables and parameters.
- It makes assumptions about the random variables, and sometimes parameters.
  - $\text{data} = \text{model} + \text{residuals}$
- Model should explain most of the variation in the data
- Residuals are a representation of **a lack-of-fit**, that is of the portion of the data unexplained by the model.

# SA – Statistical Inference

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- Estimation represents ways or a process of learning and determining the population parameter based on the model fitted to the data.
- **Point estimation** and **interval estimation**, and **hypothesis testing** are three main ways of learning about the population parameter from the sample statistic.
- There are **two** forms of estimation:
  - **Point estimation**  
A single value that estimates the parameter. Point estimates are single values calculated from the sample
  - **Confidence Intervals**  
Gives a range of values for the parameter Interval estimates are intervals within which the parameter is expected to fall, with a certain degree of confidence.

# SA – Statistical Inference



## Example (Point Estimation):

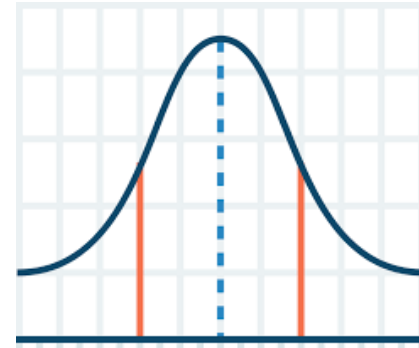
- In **point estimation**, we estimate an unknown parameter using a **single number** that is calculated from the sample data.
- Based on sample results, we estimate that  $p$ , the proportion of all U.S. adults who are in favor of stricter gun control, is **0.6**.

## Example (Interval Estimation):

- In **interval estimation**, we estimate an unknown parameter using an **interval of values** that is likely to contain the true value of that parameter (and state how confident we are that this interval indeed captures the true value of the parameter).
- Based on sample results, we are **95%** confident that  $p$ , the proportion of all U.S. adults who are in favor of stricter gun control, is between **0.57** and **0.63**.

# SA – Confidence Interval

- A confidence interval is how much uncertainty there is with any particular statistic.
- Confidence intervals are often used with a margin of error.
- It tells you how confident you can be that the results from a poll or survey reflect what you would expect to find if it were possible to **survey the entire population**.
- Confidence intervals are intrinsically connected to confidence levels.
- Essentially, a confidence interval is a **range of values** we are fairly sure our **true value** lies in.



## SA – Confidence Interval



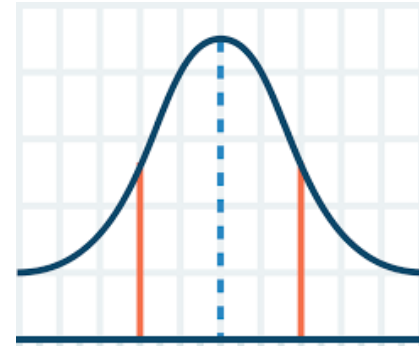
*“Confidence interval for a mean is a **range of values** that is **likely to contain a population mean** with a certain level of confidence.”*

# SA – Confidence Interval



## Example:

- Find the number of samples  $n$ , calculate the mean  $\bar{X}$  of those samples, and the standard deviation  $s$
- Using the example data:
  - Number of samples:  $n = 40$
  - Mean:  $\bar{X} = 175$
  - Standard Deviation:  $s = 20$
- Note: we should use the standard deviation of the entire **population**, but in many cases we won't know it.
- We can use the standard deviation for the **sample** if we have a large enough sample (at least  $n=30$ , hopefully more).



# SA – Confidence Interval



## Example:

- Decide what Confidence Interval we want:
  - 95% or 99% are common choices.
- Then find the "Z" value for that Confidence Interval here. ----->
- For 95% the Z value is **1.960**
- If the population standard deviation is known, use the z-distribution. If the population standard deviation is not known, use the t-distribution.

| Confidence Interval | Z     |
|---------------------|-------|
| 80%                 | 1.282 |
| 85%                 | 1.440 |
| 90%                 | 1.645 |
| 95%                 | 1.960 |
| 99%                 | 2.576 |
| 99.5%               | 2.807 |
| 99.9%               | 3.291 |

# SA – Confidence Interval

## Example:

- Use that Z in this formula for the Confidence Interval:

$$\bar{x} \pm z \frac{s}{\sqrt{n}}$$

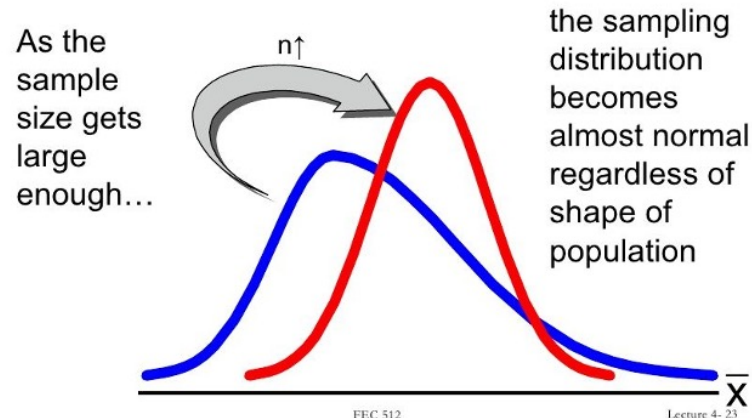
Where:-

- **X** is the mean
  - **Z** is the chosen Z-value from the table above
  - **s** is the standard deviation
  - **n** is the number of samples
- And we have:  $175 \pm 1.960 \times \frac{s}{\sqrt{n}}$
  - Which is: **175cm ± 6.20cm**
  - In other words: from **168.8cm** to **181.2cm**
  - The value after the ± is called the **margin of error**
    - The margin of error in our example is **6.20cm**



# SA – Central Limit Theorem (CLT)

- The **Central Limit Theorem (CLT)** states that for a large-enough sample size from a population with a finite level of variance, the **mean** of all samples from the same population will be approximately equal to the mean of the population.
- The larger the sample size, the better will be the normal approximation to the sampling distribution of  $\bar{x}$ .
- Rule of thumb:
  - $n > 30$
  - It can be lower!



## SA – Central Limit Theorem (CLT)



Example:

- If the population mean of number of fish caught per trip to a particular fishing hole is **3.2** and the population standard deviation is **1.8**, what are the mean and standard deviation of the sampling distribution for samples of size **40** trips?
  - Population mean = sampling mean = 3.2
  - Population std,  $\sigma = 1.8$
  - Sampling std =  $\frac{\sigma}{\sqrt{n}} = \frac{1.8}{\sqrt{40}} = 0.285$

## SA – Confidence Interval ( Python using Scipy )



- If the sample ( $n < 30$ ), we can use the *t.interval()* function from scipy.
- Example shows how to calculate a confidence interval for the true population mean **height (in inches) of a certain species of plant**, using a sample of 15 plants:

```
data = [12, 12, 13, 13, 15, 16, 17, 22, 23, 25, 26, 27, 28, 28, 29]
```

## SA – Confidence Interval ( Python using Scipy )

```
import numpy as np
import scipy.stats as st

#define sample data
data = [12, 12, 13, 13, 15, 16, 17, 22, 23, 25, 26, 27, 28, 28, 29]

#create 95% confidence interval for population mean weight
st.t.interval(confidence=0.95, df=len(data)-1, loc=np.mean(data), scale=st.sem(data))

(16.75776979778498, 24.042230202215016)
```

- The 95% confidence interval for the true population mean height is (16.758, 24.042).

## SA – Confidence Interval ( Python using Scipy )

- You'll notice that the larger the confidence level, the wider the confidence interval. Calculate a 99% C.I. for the exact same data:

```
import numpy as np
import scipy.stats as st

#define sample data
data = [12, 12, 13, 13, 15, 16, 17, 22, 23, 25, 26, 27, 28, 28, 29]

#create 95% confidence interval for population mean weight
st.t.interval(confidence=0.99, df=len(data)-1, loc=np.mean(data), scale=st.sem(data))

(15.3447902793728, 25.455209720627195)
```

- The 99% confidence interval for the true population mean height is (15.348, 25.455). Notice that this interval is wider than the previous 95% confidence interval.

# SA – Hypothesis Testing



- Hypothesis testing is a statistical test that is used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population.
- It examines two opposing hypotheses about a population:
  - **The null hypothesis ( $H_0$ , H-null)**
    - Being used to show there is no relationship between two measured phenomena, or no association among group.
    - It is generally assumed to be true until evidence indicates otherwise
  - **The alternative hypothesis**
    - Is contrary to the null hypothesis. It is usually taken to be that the observations are the result of a real effect

## SA – Hypothesis Testing ( p-value )

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- A p value is used in hypothesis testing to help you support or reject the null hypothesis.
- The p value is the **evidence against** a null hypothesis.
- The **smaller** the p-value, the **stronger** the evidence that you should **reject** the null hypothesis.
- Can be thought of as the  **$P(\text{Rejecting } H_0 \mid H_0 \text{ is true})$**
- P values are expressed as decimals although it may be easier to understand what they are if you convert them to a percentage.

# SA – Hypothesis Testing ( t-test )

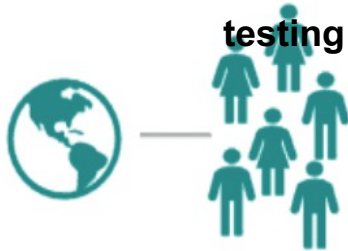
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- A *t*-test (also known as Student's *t*-test) is a tool for **evaluating the means of one or two populations** using hypothesis testing.
- A *t*-test may be used to evaluate whether:
  - A **single group** differs from a known value (a one-sample *t*-test),
  - **Two groups** differ from each other (an independent two-sample *t*-test),
  - There is a significant difference in **paired measurements** (a paired, or dependent samples *t*-test).



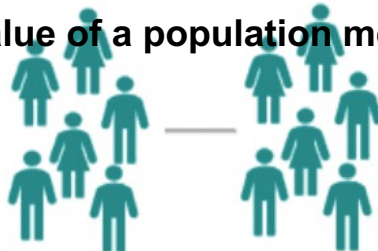
# SA – Hypothesis Testing ( t-test )

## One sample t-Test



Is there a **difference** between a **group** and the **population**

## Independent samples t-Test



Is there a **difference** between **two groups**

## Paired samples t-Test



Is there a **difference** in a **group** between **two points in time**

testing the value of a population mean

## SA – Hypothesis Testing ( 1 sample t-test )

- Testing the value of a population mean
- We can also use Scipy for this. Lets use “*brain\_data.csv*”

```
import pandas as pd
import scipy as sc

brain = pd.read_csv('data/brain_size.csv', sep=';')
sc.stats.ttest_1samp(brain['VIQ'], 0 )
```

```
TtestResult(statistic=30.08809997084933, pvalue=1.328919646872788e-28, df=39)
```

- With a p-value of  $10^{-28}$  we can claim that the population mean for the IQ (VIQ measure) is **not 0**.

## SA – Hypothesis Testing ( 2-sample t-test )

- Testing for difference across populations
- We have seen above that the mean VIQ in the male and female populations were different. But it is significant?
- To test if this is significant, we do a 2-sample t-test with *scipy.stats.ttest\_ind()*

```
female_viq = brain[brain['Gender'] == 'Female']['VIQ']  
male_viq = brain[brain['Gender'] == 'Male']['VIQ']  
print('Female viq mean:', female_viq.mean())  
print('Male viq mean:', male_viq.mean())  
sc.stats.ttest_ind(female_viq, male_viq)
```

```
Female viq mean: 109.45
```

```
Male viq mean: 115.25
```

```
TtestResult(statistic=-0.7726161723275012, pvalue=0.44452876778583217, df=38.0)
```

## SA – Hypothesis Testing ( Paired t-test )

---

- Repeated measurements on the same individuals
- From the data, PIQ, VIQ, and FSIQ give 3 measures of IQ.
- Let's test if FSIQ and PIQ are significantly different. We can use a paired test

```
sc.stats.ttest_rel(brain['FSIQ'], brain['PIQ'])
```

```
TtestResult(statistic=1.7842019405859857, pvalue=0.08217263818364236, df=39)
```

# SA – Hypothesis Testing ( Errors )



- When we conduct a hypothesis test there a couple of things that could go wrong.
- There are two kinds of errors, which by design cannot be avoided, and we must be aware that these errors exist.
- Briefly:
  - **Type 1** errors happen when we **reject a true null hypothesis**
  - **Type 2** errors happen when we **fail to reject a false null hypothesis**
- In an ideal world we would always reject the null hypothesis when it is false, and we would not reject the null hypothesis when it is indeed true.
- But there are two other scenarios that are possible, each will result in an error.

# SA – Hypothesis Testing ( Errors )

---

## Type I Error



## Type II Error



# SA – Hypothesis Testing ( Type 1 Errors )



- The first kind of error that is possible involves the rejection of a null hypothesis that is actually true.
- This kind of error is called a type I error, and is sometimes called an error of the first kind.
- **Type I** errors are equivalent to false positives.
- Example:
  - A drug being used to treat a disease.
  - If we reject the null hypothesis in this situation, then our claim is that the drug does in fact have some effect on a disease.
  - But if the null hypothesis is true, then in reality the drug does not combat the disease at all.
  - The drug is falsely claimed to have a positive effect on a disease.

## SA – Hypothesis Testing ( Type 1 Errors )



- Type 1 errors can be controlled.
- The value of alpha, which is related to the level of significance that we selected has a direct bearing on type I errors.
- Alpha is the maximum probability that we have a type I error.
- For a 95% confidence level, the value of alpha is 0.05.
- This means that there is a 5% probability that we will reject a true null hypothesis.
- In the long run, one out of every twenty hypothesis tests that we perform at this level will result in a type I error.



## SA – Hypothesis Testing ( Type 2 Errors )



- The other kind of error that is possible occurs when we do not reject a null hypothesis that is false.
- This sort of error is called a type 2 error, and is also referred to as an error of the second kind.
- Type II errors are equivalent to false negatives.
- Example:
  - A drug being used to treat a disease.
  - A type II error would occur if we accepted that the drug had no effect on a disease, but in reality it did.
- The probability of a type II error is given by the Greek letter beta.

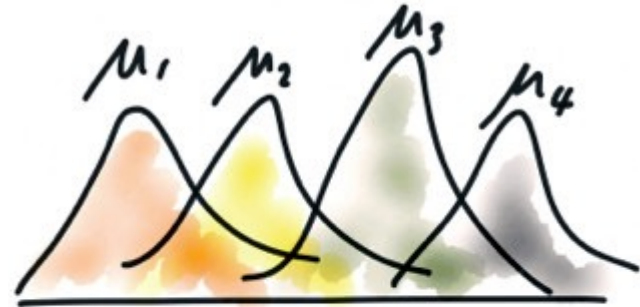
# SA – Hypothesis Testing ( How to Avoid Errors )



- Type I and type II errors are part of the process of hypothesis testing.
- Although the errors cannot be completely eliminated, we can **minimize** one type of error.
- Typically when we try to decrease the probability one type of error:
  - The probability for the other type increases.
  - We could decrease the value of alpha from 0.05 to 0.01, corresponding to a 99% level of confidence.
  - However, if everything else remains the same, then the probability of a type II error will nearly always increase.
- Many times the real world application of our hypothesis test will determine if we are more accepting of type I or type II errors.
- This will then be used when we design our statistical experiment.

# SA – One-Way Analysis of Variance (ANOVA)

- Used to determine whether there are any statistically significant differences between the means of two or more independent (unrelated) groups
- Mostly being used for a minimum of three and more, rather than two groups.



ANOVA

$$\mu_1 = \mu_2 = \mu_3 = \mu_4 ?$$

# SA – One-Way Analysis of Variance (ANOVA)



- A one-way ANOVA has the below given null and alternative hypotheses:
  - $H_0$  (null hypothesis):  $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$  (It implies that the means of all the population are equal)
  - $H_1$  (null hypothesis): It states that there will be at least one population mean that differs from the rest

# SA – One-Way Analysis of Variance (ANOVA)



- Scenario:
- Researchers took 20 cars of the same to take part in a study. These cars are randomly doped with one of the four-engine oils and allowed to run freely for 100 kilometers each. At the end of the journey, the performance of each of the cars is noted.
- Data:
  - # Performance when each of the engine oil is applied
  - performance1 = [89, 89, 88, 78, 79]
  - performance2 = [93, 92, 94, 89, 88]
  - performance3 = [89, 88, 89, 93, 90]
  - performance4 = [81, 78, 81, 92, 82]

## SA – One-Way Analysis of Variance (ANOVA)

- Using scipy, we conduct one-way ANOVA using the `f_oneway()` function.

```
import scipy as sc

performance1 = [89, 89, 88, 78, 79]
performance2 = [93, 92, 94, 89, 88]
performance3 = [89, 88, 89, 93, 90]
performance4 = [81, 78, 81, 92, 82]

sc.stats.f_oneway(performance1, performance2, performance3, performance4)

F_onewayResult(statistic=4.625000000000002, pvalue=0.01633645983978022)
```

## SA – One-Way Analysis of Variance (ANOVA)



- The F statistic and p-value turn out to be equal to 4.625 and 0.016336498 respectively.
- Since the **p-value is less than 0.05** hence we would **reject the null hypothesis**.
- This implies that **we have sufficient proof** to say that **there exists a difference** in the performance among four different engine oils.

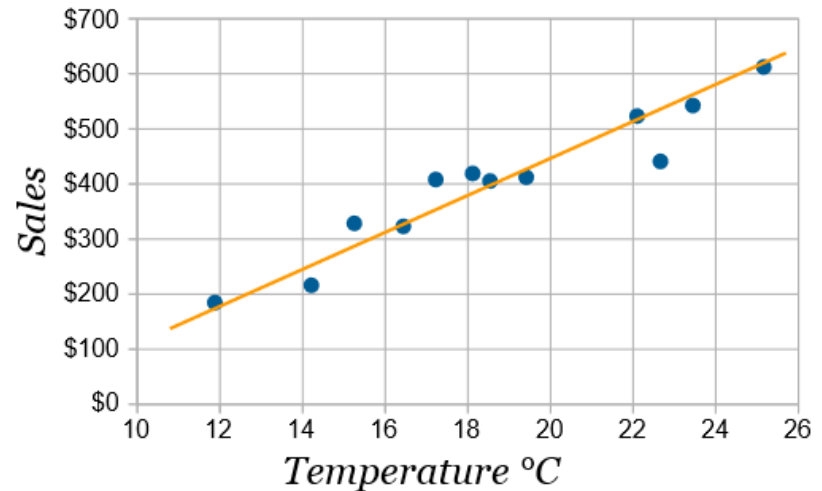
Like what Ryan Gosling would say to you.....





# SA – Least Squares Regression (Linear Regression)

- A simple linear regression model that describes the relationship between two variables  $x$  and  $y$ .
- It is also called as the **Line of best fit**.
- We can place the line "by eye": try to have the line as close as possible to all points, and a similar number of points above and below the line.



# SA – Least Squares Regression (Linear Regression)



The line:

- Our aim is to calculate the values **m** (slope) and **b** (y-intercept) in the equation of a line:

$$y = mx + b$$

- Where:
  - **y** = how far up
  - **x** = how far along
  - **m** = Slope or Gradient (how steep the line is)
  - **b** = the Y Intercept (where the line crosses the Y axis)

# SA – Least Squares Regression (Linear Regression)



## Example:

- Sam found how many hours of sunshine vs how many ice creams were sold at the shop from Monday to Friday:

| "x"<br>Hours of Sunshine | "y"<br>Ice Creams Sold |
|--------------------------|------------------------|
| 2                        | 4                      |
| 3                        | 5                      |
| 5                        | 7                      |
| 7                        | 10                     |
| 9                        | 15                     |

- Find the best **m** (slope) and **b** (y-intercept) that suits that data.

# SA – Least Squares Regression (Linear Regression)



## Example:

- Step 1: For each (x,y) calculate  $x^2$  and  $xy$ . Sum all of them:
  - Also  $N$  (number of data values/points) = 5

| x              | y              | $x^2$             | $xy$             |
|----------------|----------------|-------------------|------------------|
| 2              | 4              | 4                 | 8                |
| 3              | 5              | 9                 | 15               |
| 5              | 7              | 25                | 35               |
| 7              | 10             | 49                | 70               |
| 9              | 15             | 81                | 135              |
| $\Sigma x: 26$ | $\Sigma y: 41$ | $\Sigma x^2: 168$ | $\Sigma xy: 263$ |

# SA – Least Squares Regression (Linear Regression)

## Example:

- Step 2: Calculate the slope,  $m$ .

$$m = \frac{(N \sum xy - \sum x \sum y)}{N(\sum x^2) - (\sum x)^2}$$

$$\triangleright = (5 \times 263 - 26 \times 41) \div (5 \times 168 - 26^2)$$

$$\triangleright = (1315 - 1066) \div (840 - 676)$$

$$\triangleright = \frac{249}{164} = 1.5183$$

- Step 3: Calculate the y-intercept,  $b$ .

$$b = \frac{\sum y - m(\sum x)}{N}$$

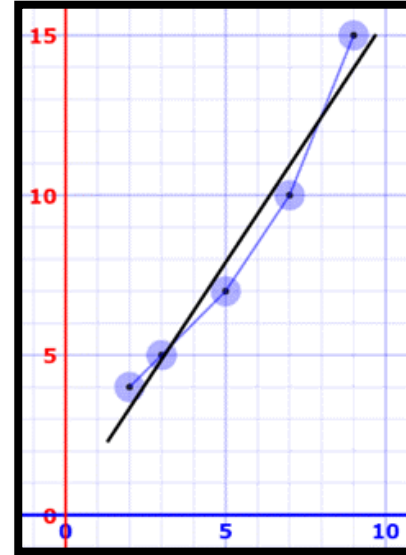
$$\triangleright = \frac{41 - 1.5183 \times 26}{5} = 0.3049$$

# SA – Least Squares Regression (Linear Regression)

## Example:

- Step 4: The equation of line
  - $y = mx + b$
  - $y = 1.518x + 0.305$

| x | y  | $y = 1.518x + 0.305$ | error |
|---|----|----------------------|-------|
| 2 | 4  | 3.34                 | -0.66 |
| 3 | 5  | 4.86                 | -0.14 |
| 5 | 7  | 7.89                 | 0.89  |
| 7 | 10 | 10.93                | 0.93  |
| 9 | 15 | 13.97                | -1.03 |



- The points would be mapped into the equation and the graph.
- Quite a nice fit.
- Sam hears the weather forecast which says "we expect 8 hours of sun tomorrow", so he uses the above equation to estimate that he will sell:
  - $y = 1.518 \times 8 + 0.305 = 12.45$  Ice Creams

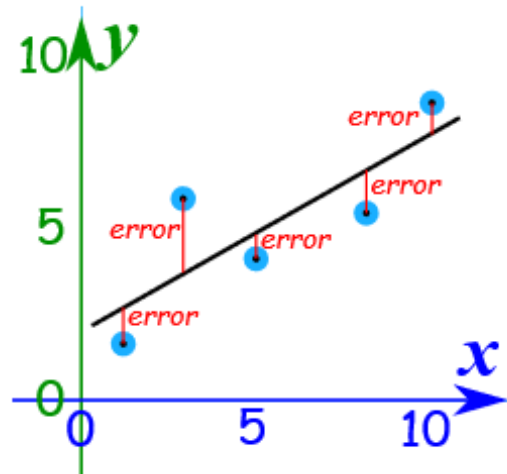
# SA – Least Squares Regression (Linear Regression)

## How it works:

- It works by making the total of the **square of the errors** as small as possible (that is why it is called "least squares").
- The straight line minimizes the sum of squared errors.
- So, when we square each of those errors and add them all up, the total is as small as possible.

## Outliers

- Least squares are sensitive to outliers.
- A strange value will pull the line towards it.



# SA – Least Squares Regression ( OLS in Python )

- We use the statsmodels package. (pip install statsmodels)
- x (independent variable) = Brain\_weight
- y (dependent variable) = Head\_size
- How to write:
  - "  $y \sim x$  "

```
import pandas as pd
import statsmodels.formula.api as smf

df = pd.read_csv('../Lectures/data/headbrain.csv')

# fitting the model
df.columns = ['Head_size', 'Brain_weight']
model = smf.ols(formula='Head_size ~ Brain_weight',
                 data=df).fit()

# model summary
print(model.summary())
```



# SA – Least Squares Regression ( OLS in Python )

| OLS Regression Results                                                                                                                |                  |                     |        |          |         |         |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|--------|----------|---------|---------|
| Dep. Variable:                                                                                                                        | Head_size        | R-squared:          |        | 0.639    |         |         |
| Model:                                                                                                                                | OLS              | Adj. R-squared:     |        | 0.638    |         |         |
| Method:                                                                                                                               | Least Squares    | F-statistic:        |        | 416.5    |         |         |
| Date:                                                                                                                                 | Tue, 02 Apr 2024 | Prob (F-statistic): |        | 5.96e-54 |         |         |
| Time:                                                                                                                                 | 11:18:32         | Log-Likelihood:     |        | -1613.4  |         |         |
| No. Observations:                                                                                                                     | 237              | AIC:                |        | 3231.    |         |         |
| Df Residuals:                                                                                                                         | 235              | BIC:                |        | 3238.    |         |         |
| Df Model:                                                                                                                             | 1                |                     |        |          |         |         |
| Covariance Type:                                                                                                                      | nonrobust        |                     |        |          |         |         |
|                                                                                                                                       | coef             | std err             | t      | P> t     | [0.025  | 0.975]  |
| Intercept                                                                                                                             | 520.6101         | 153.215             | 3.398  | 0.001    | 218.759 | 822.461 |
| Brain_weight                                                                                                                          | 2.4269           | 0.119               | 20.409 | 0.000    | 2.193   | 2.661   |
| Omnibus:                                                                                                                              | 2.687            | Durbin-Watson:      |        | 1.726    |         |         |
| Prob(Omnibus):                                                                                                                        | 0.261            | Jarque-Bera (JB):   |        | 2.321    |         |         |
| Skew:                                                                                                                                 | 0.207            | Prob(JB):           |        | 0.313    |         |         |
| Kurtosis:                                                                                                                             | 3.252            | Cond. No.           |        | 1.38e+04 |         |         |
| Notes:                                                                                                                                |                  |                     |        |          |         |         |
| [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.                                           |                  |                     |        |          |         |         |
| [2] The condition number is large, 1.38e+04. This might indicate that there are strong multicollinearity or other numerical problems. |                  |                     |        |          |         |         |

# SA – Least Squares Regression ( OLS in Python )



- **Description of some of the terms in the table :**
- **R-squared value:**
  - The R-squared value ranges from 0 to 1.
  - An R-squared of 100% indicates that changes in the independent variable completely explain all changes in the dependent variable (s).
  - If the r-squared value is 1, it indicates a perfect fit.
  - The r-squared value in our example is **0.638**.
- **F-statistic:**
  - The F-statistic compares the combined effect of all variables. Simply put, if your alpha (*usually 0.05 -> 95% confidence interval*) level is greater than your p-value, you should reject the null hypothesis.
- **coef:**
  - the coefficients of the regression equation's independent variables.

## SA – Least Squares Regression ( OLS in Python )



- If we use 0.05 as our significance level, we reject the null hypothesis and accept the alternative hypothesis as  $p < 0.05$ .
- As a result, we can conclude that there is a relation between head size and brain weight.

# SA – Least Squares Regression ( OLS in Python )

```
import matplotlib.pyplot as plt
import numpy as np

df = df.sort_values(['Brain_weight'])

b,a = np.polyfit(df['Brain_weight'],
                df['Head_size'],
                deg=1)

plt.scatter(df['Brain_weight'],
            df['Head_size'])
plt.plot(df['Brain_weight'], a + b * df['Brain_weight'])
plt.xlabel('Brain Weight')
plt.ylabel('Head size')
```

